

Calendar — Fast Tricks & Shortcuts

Anchor-Day Method, Month Codes, and Every Speed Trick You Need

Aptitude Grind

This file collects the **fast calculation methods** — the actual shortcuts you execute under exam pressure. For the full catalogue of question *types* that use these tricks, see `03-question-pattern-catalog.tex`.

1 The Anchor-Day Method (Fastest for Placement Tests)

Instead of counting odd days from year 1 (or even using the mod-400 shortcut from `01-fundamentals.tex`), work relative to a **known reference date** close to your target date. Most questions either give you a reference day directly, or you can use a memorized anchor.

Useful Anchors to Memorize

- 1 January 2000 → **Saturday** (most practical anchor for modern-date questions)
- 1 January 1900 → **Monday**
- 1 January 2100 → **Friday**

Anchor-Day Procedure

1. Find the number of complete years between the anchor and target year.
2. Odd days contributed = (ordinary years $\times$ 1) + (leap years $\times$ 2), reduced mod 7.
3. Add odd days for the months elapsed in the target year, then add the date number.
4. Add this to the anchor's day-of-week position (mod 7) and read off the result.

Why This Beats the Mod-400 Method

The mod-400 method is for when you're asked about an arbitrary date, possibly centuries away. The anchor method is for when your target date is within a normal human lifetime of a known reference point — which is nearly every placement-test question. Fewer years to sum = fewer chances to make an arithmetic slip.

Worked Example

If 1 Jan 2000 was a Saturday, what day was 1 Jan 2010?

- Years 2000–2009 (10 years elapsed up to 1 Jan 2010)
- Leap years: 2000, 2004, 2008 → 3 leap, 7 ordinary

- Odd days = $(3 \times 2) + (7 \times 1) = 13 \equiv 6 \pmod{7}$
- Saturday +6 → **Friday**

2 Month Codes Method (a.k.a. Doomsday-style Shortcut)

A faster alternative used heavily in coaching materials: assign each month a fixed **code** representing its odd-day contribution from 1 January of a non-leap year up to the 1st of that month.

Month Codes — Non-Leap Year

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	0	3	3	6	1	4	6	2	5	0	3	5

Leap Year Adjustment

For a **leap year**, subtract 1 from the codes of **January and February only** (the extra day, 29 Feb, hasn't occurred yet when computing odd days up to those months). From March onward in a leap year, use the codes as-is — the leap day has already been absorbed.

$$\text{Day code} = (\text{Date} + \text{Month code} + \text{Year's odd days from anchor}) \pmod{7}$$

Match the result against the Odd-Days → Day table in `01-fundamentals.tex`.

When to Use Which Method

The odd-days-from-scratch method is more intuitive and safer under exam pressure because you derive everything instead of recalling a code table. The month-code method is faster once memorized, but a single misremembered code silently gives a wrong answer. For placement tests with negative marking, prefer the method you can execute without doubt — practice both, then pick one.

3 Day After / Before N Days — Pure Modulo, No Calendar Needed

This is the single fastest calculation in the entire topic, and it needs **no leap-year knowledge at all** when you're just counting raw days forward or backward from a *named* day (not a date).

Rule

$N \pmod{7}$ tells you how many weekday-steps to move. Forward for “after,” backward for “before.”

Example — After

Today is Tuesday. What day after 100 days?
 $100 \bmod 7 = 2 \rightarrow \text{Tuesday} + 2 \rightarrow \textbf{Thursday}$

Example — Before

100 days before Friday.
 $100 \bmod 7 = 2 \rightarrow \text{Friday} - 2 \rightarrow \textbf{Wednesday}$

Common Mistake

Reaching for the odd-days-per-year method here. If the question only gives you a day name (not a calendar date) and a day count, you never need leap years — just $N \bmod 7$. Leap years only matter once you're crossing actual calendar dates, because then the *number of days in between* depends on whether 29 February falls inside that span.

4 Leap Year — Quick Check and Quick Count

Quick Check for One Year

Divisible by 4? No $\rightarrow$ Not leap. Yes $\rightarrow$ century year (ends in 00)? No $\rightarrow$ Leap. Yes $\rightarrow$ divisible by 400? Yes $\rightarrow$ Leap; No $\rightarrow$ Not leap.

Quick Count for a Range

Leap years from 1 to $N = \lfloor N/4 \rfloor - \lfloor N/100 \rfloor + \lfloor N/400 \rfloor$.
 For a range $[A, B]$: $\text{count}(B) - \text{count}(A - 1)$.

Worked Example — Leap Years From 1601 to 2000

$$\begin{aligned}\text{count}(2000) &= \lfloor 2000/4 \rfloor - \lfloor 2000/100 \rfloor + \lfloor 2000/400 \rfloor = 500 - 20 + 5 = 485 \\ \text{count}(1600) &= \lfloor 1600/4 \rfloor - \lfloor 1600/100 \rfloor + \lfloor 1600/400 \rfloor = 400 - 16 + 4 = 388 \\ \text{Leap years in } [1601, 2000] &= 485 - 388 = \mathbf{97}\end{aligned}$$

(Verified independently with a direct year-by-year count.)

5 Weekday Frequency Inside a Month

Every month has at least 4 of each weekday. Whether a weekday occurs a 5th time depends purely on **how many days the month has beyond 4 full weeks (28 days)**, and **which weekday the month starts on**.

Rule

Let $L = (\text{days in month}) - 28$. The **first L weekdays** starting from the month's 1st-day weekday occur **5 times**; every other weekday occurs exactly 4 times.

Month length	L	Weekdays occurring 5 times
28 days (Feb, non-leap)	0	None — every weekday occurs exactly 4 times
29 days (Feb, leap)	1	Only the weekday the month starts on
30 days	2	The starting weekday and the next one
31 days	3	The starting weekday and the next two

Worked Example

A 31-day month starts on Tuesday. Which weekdays occur 5 times?

$L = 31 - 28 = 3 \rightarrow$ first 3 weekdays from Tuesday: **Tue, Wed, Thu**

(Verified with a full day-by-day count for a real 31-day month starting on a Monday — Mon/Tue/Wed each appeared 5 times, matching the rule.)

6 Month Start/End Day — Jump Straight From One to the Other**Rule**

To go from the weekday of the 1st to the weekday of any day D of the same month, add $(D - 1) \bmod 7$ to the 1st's weekday.

Worked Example — Start to End

1 June = Wednesday. What day is 30 June?

$(30 - 1) \bmod 7 = 29 \bmod 7 = 1 \rightarrow \text{Wednesday} + 1 \rightarrow \text{Thursday}$

Worked Example — Start of Next Month

1 March = Monday. Find 1 April. (March has 31 days.)

$31 \bmod 7 = 3 \rightarrow \text{Monday} + 3 \rightarrow \text{Thursday}$

This chains naturally: keep adding (month length mod 7) to hop from one month's 1st to the next.

7 Same-Calendar-Year Gaps — Corrected

Coaching material widely repeats “an ordinary year repeats after 6, 11, **17**, or 28 years.” This was checked directly by scanning three thousand years of real calendars, and **17 never actually occurs**. Here's what's actually true:

Starting year type	Typical gap	Rare gap (broken century boundary)
Ordinary	6 or 11 years	12 years
Leap	28 years	40 years

Don't Memorize "17"

It's a commonly repeated number in Indian test-prep material that does not hold up when checked against actual calendars. If you see it in another source, verify with the running-odd-day-total method below instead of trusting it blindly.

Running-Odd-Day-Total Method — Always Correct

1. Starting from your given year, add 1 odd day for each ordinary year that passes and 2 for each leap year, keeping a running total mod 7.
2. Stop at the first year where the running total hits 0 **and** the year's leap/ordinary type matches the starting year's type (same weekday for 1 Jan isn't enough if one year is leap and the other isn't, since their Februaries differ).

8 Quick Reference — Which Trick for Which Situation

Situation	Use
Target date centuries away from any known anchor	Mod-400 method (Notes 01)
Target date within ~100–150 years of a known anchor	Anchor-Day method (§1)
You've memorized the month-code table	Month-code method (§2)
Only a day name + a day count given (no calendar date)	$N \bmod 7$ (§3)
"How many leap years between X and Y"	Leap-count formula (§4)
"Which weekday(s) occur 5 times"	Weekday-frequency rule (§5)
Need another day's weekday within the same month	Start/end day rule (§6)
"Same calendar as year X repeats in which year"	Running-odd-day-total (§7) — don't guess 17